Hand Gesture Recognition Using Convolutional Neural Networks (CNN)

Abstract

A significant use of deep learning and computer vision is hand gesture recognition, which allows computers to comprehend and interpret human hand movements. In order to categorise hand motions into five groups—Backward, Forward, Left, Right, and Unknown Convolutional Neural Network (CNN) was created for this research. The roughly 13,000 photos in the dataset were separated into training and validation datasets after being preprocessed. TensorFlow and Keras were used to train the model, which attained a validation accuracy of roughly 99.15%. The findings show that CNNs are capable of very accurate image categorisation and efficient gesture pattern learning.

1. Introduction

In human-computer interaction systems, hand gesture recognition is crucial. Instead of using conventional input devices, it enables users to interact with machines using their hands. Robotics, virtual reality, gaming systems, smart homes, and assistive technology are some of the uses for gesture recognition.

For picture classification tasks, deep learning methods in particular, Convolutional Neural Networks (CNNs) have shown great efficacy. CNNs can automatically extract visual features from images without the need for human feature extraction. The goal of this research is to develop a CNN-based hand gesture detection system that can categorise pictures into pre-established gesture groups.

2. Objectives

The project's goals are:

- To create a CNN model for classifying hand gestures.
- To prepare picture datasets for training by preprocessing them.
- To use Keras and TensorFlow to train and validate the CNN model.
- To assess the accuracy, loss, confusion matrix, and classification metrics of the model.
- To attain excellent categorisation accuracy in every class of gestures.

3. Dataset Description

The Hand Gesture Dataset, which comprises over 13,000 photos categorised into five types, was utilised in this project:

- In reverse
- Proceed
- On the left
- Correct
- Unknown

Based on their class labels, the photos are kept in different directories. TensorFlow can automatically assign labels during dataset loading thanks to this structure. The dataset's diversity of hand positions and orientations contributes to the model's increased robustness.

An 80:20 ratio was used to split the dataset into subsets for training and validation. 2,600 photos were utilised for validation and about 10,400 images were used for training.

4. Data Preprocessing

Prior to CNN model training, data pretreatment was carried out.

The preprocessing procedures listed below were used:

- TensorFlow's image_dataset_from_directory() method was used to load the images.* Based on folder names, labels were automatically assigned.
- Pictures were downsized to 128 by 128 pixels.
- The 80:20 ratio was used to divide the dataset into training and validation sets.* A Rescaling layer was used to normalise pixel values and transform them into the range [0,1].

Model performance and training stability were enhanced by these preprocessing methods.

5. CNN Architecture

A Convolutional Neural Network (CNN) architecture was developed for gesture classification.

The architecture contains:

- Conv2D Layer (32 filters, 3x3 kernel with ReLU activation)

- MaxPooling2D Layer.
- Conv2D Layer (64 filters, 3x3 kernel with ReLU activation)
- MaxPooling2D Layer.
- Conv2D Layer (128 filters, 3x3 kernel, ReLU activation).
- MaxPooling2D Layer.
- Flatten Layer.
- Dense Layer (128 neurones with ReLU activation)
- Dropout Layer (0.5).
- Output Dense Layer (5 neurones with Softmax activation)

The model has over 3.3 million trainable parameters.

6. Model Training

The CNN model was trained with the following settings:

- Optimiser: Adam* Loss Function: Sparse Categorical Cross-Entropy
- The metrics are: Accuracy* Epochs: ten.
- Batch size: 32.

The model was trained on the training dataset and then tested on the validation dataset after each epoch.

7. Results and Discussion

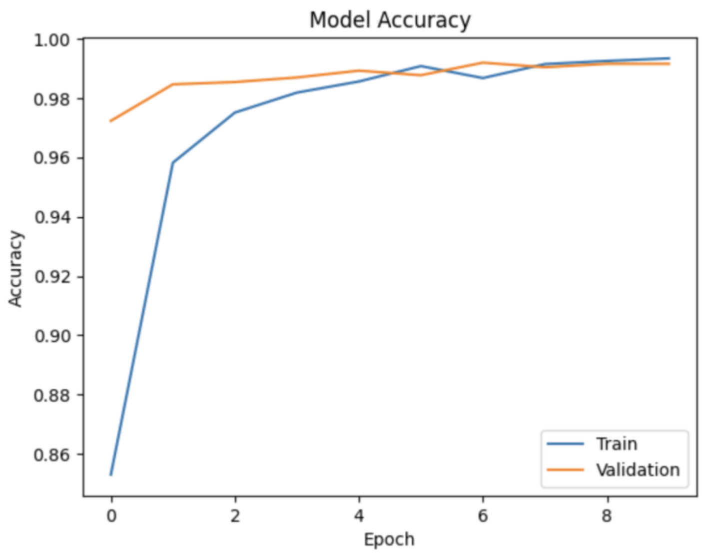
The model performed exceptionally well on the validation dataset.

Validation Performance

- Validation accuracy is 99.15%.
- Validation loss = 0.0339.

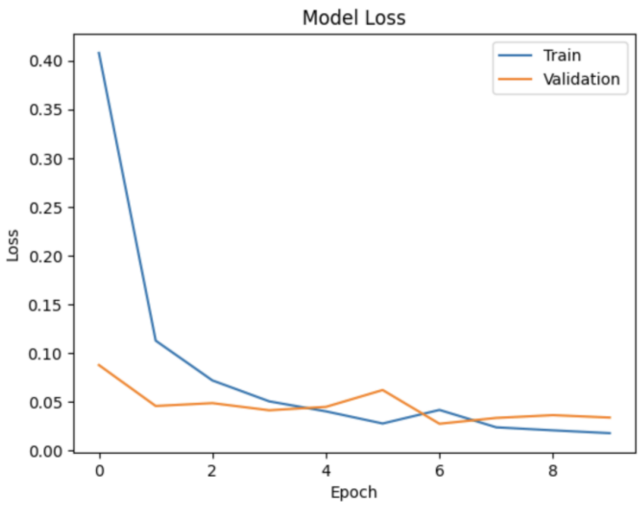
Accuracy Curve

Training and validation accuracy rose continuously throughout training and converged at about 99%, indicating excellent learning and good generalisation.



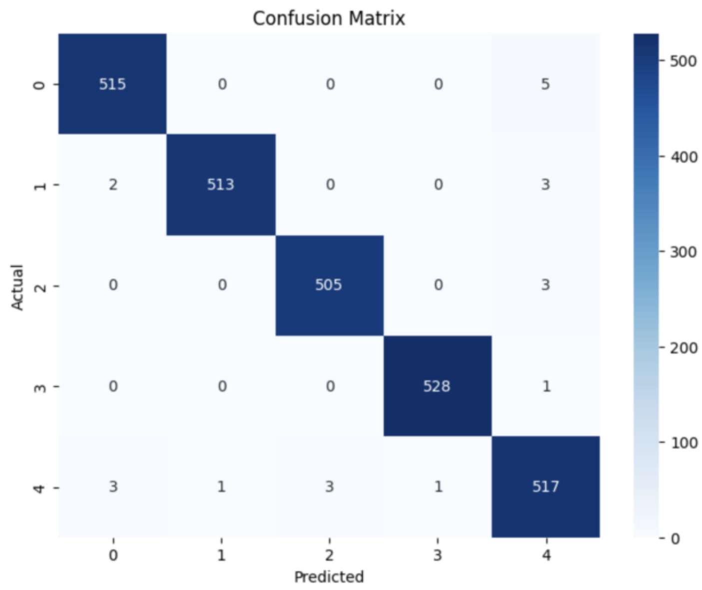
Loss Curve

The training and validation loss dropped consistently over the training epochs, indicating good optimisation and convergence.



Confusion Matrix

The confusion matrix reveals that the majority of predictions fall along the diagonal, indicating that gesture classes are correctly classified with few misclassifications.



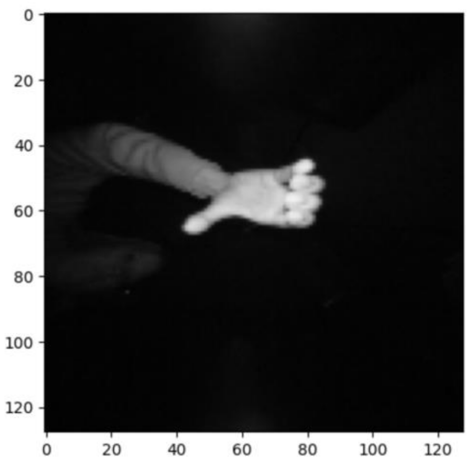
Classification Report

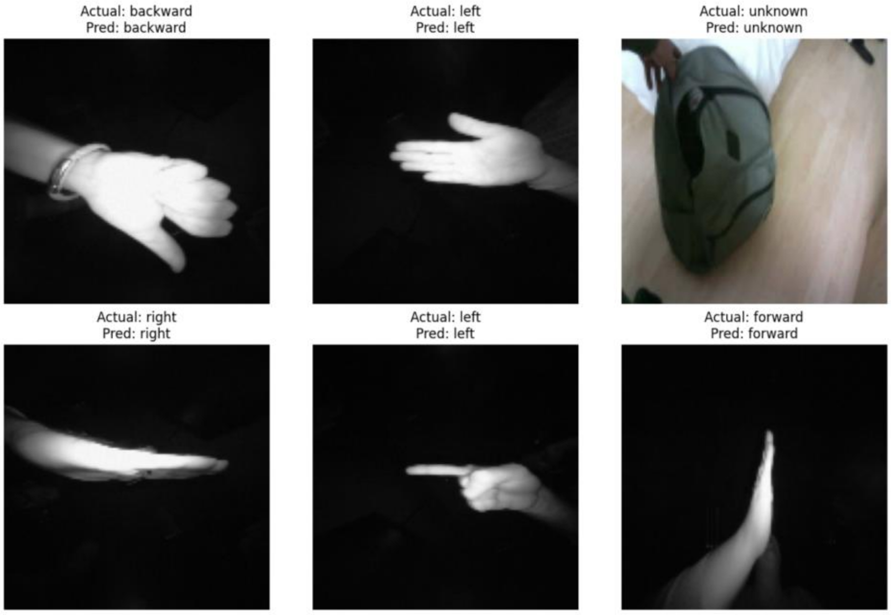
The categorisation report achieved precision, recall, and F1-scores close to 0.99 across all gesture categories, indicating balanced and highly accurate performance.

	precision	recall	f1-score	support
backward	0.99	0.99	0.99	520
forward	1.00	0.99	0.99	518
left	0.99	0.99	0.99	508
right	1.00	1.00	1.00	529
unknown	0.98	0.98	0.98	525
accuracy			0.99	2600
macro avg	0.99	0.99	0.99	2600
weighted avg	0.99	0.99	0.99	2600

Sample Predictions:

The model properly categorised several hand gesture photos from the validation dataset. Sample predictions show how well the trained CNN model performs.





Conclusion.

In this study, we successfully created a Convolutional Neural Network (CNN) for hand gesture identification. The model was trained using about 13,000 photos from five gesture classes and obtained a validation accuracy of 99.15%.

The confusion matrix and classification report show that the model performs exceptionally well across all classes, with few classification errors. The findings show that CNN-based techniques are particularly effective in gesture recognition applications.

Future Work

Future upgrades could include:

- Real-time gesture recognition with camera input.
- Launching as a web or mobile app.
- Testing on larger and more diverse datasets.
- Using transfer learning approaches to boost performance.
- Supporting more gesture classes.

References

1. Tensorflow documentation.
2. Keras documentation.
3. Hand Gestures Dataset (Kaggle).
4. Goodfellow, I., Bengio, Y., and Courville, A. Deep learning.